

Capital Bikeshare Hourly Demand: Exploratory Data Analysis Report

STAT 24620 / FINM 34700 / STAT 32950 — Group Project (Draft)

Spring 2026

1 Introduction and research question

We work with the *Capital Bikeshare hourly demand* dataset (`bike_sharing_hourly.csv`, UCI Machine Learning Repository). It records every hour of operation of the Washington D.C. bikeshare system between 2011-01-01 and 2012-12-31, together with calendar and weather covariates. The target is `cnt`, the total number of rentals during the hour.

Tentative research question. *What drives hourly bike-rental demand in Washington D.C., and how do calendar and weather variables jointly shape it?* Concretely: (i) discover the dominant patterns in hourly demand using a multivariate-structure method (PCA / clustering) and (ii) quantify and predict demand using a supervised method (regression / random forest), then compare the two views.

Data at a glance. The CSV has $n = 17,379$ rows and 18 columns with *no missing values*. Columns: calendar (`dteday`, `season`, `yr`, `mnth`, `hr`, `holiday`, `weekday`, `workingday`), weather (`weathersit`, `temp`, `atemp`, `hum`, `windspeed`), and counts (`casual`, `registered`, `cnt`). By definition $\text{cnt} = \text{casual} + \text{registered}$, so the two components may not be used as predictors of `cnt`. A derived binary label `high_demand` flags hours above the sample median.

Two minor facts inform downstream choices. First, the file actually contains $17,379 < 17,544 = 731 \times 24$ rows: **165 day-hours are absent from the table**, which we treat as zero-rental hours (consistent with the `cnt` minimum of 1). Second, the weather variables are pre-normalized to $[0, 1]$ by the UCI provider; we de-normalize them only for plotting.

This report has two parts. **Part 1** (§§2–6) is the classical descriptive EDA: distributions, time series, hour-of-day patterns, weather and calendar effects, correlation, and the binary label. **Part 2** (§§7–12) is a deeper multivariate-structure exploration: mutual information, temporal dependence, variance decomposition, PCA / K-means / hierarchical / t-SNE on day-level 24-hour profiles, hour $\times$ temperature interaction surface, and Mahalanobis multivariate outliers.

Part 1 — Classical descriptive EDA

2 Target variable, time series, and basic patterns

Marginal distribution. Hourly `cnt` ranges from 1 to 977 with mean 189.5 and sd 181.4 (median 142). The raw distribution is strongly right-skewed (skewness = 1.28, excess kurtosis = 1.42;

Figure 1, left). After $\log(1 + \text{cnt})$ the skewness drops to -0.82 , but the distribution becomes clearly *bimodal* (right) — a hint of two latent demand regimes that Part 2 will recover quantitatively.

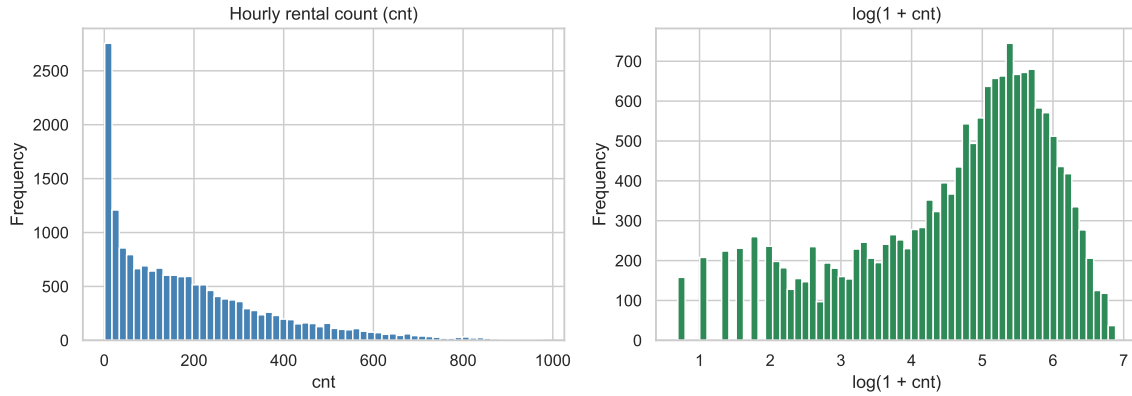


Figure 1: Raw vs. log-transformed hourly demand.

Long-term trend and seasonality. Two years of data (2011–2012) show three time-scales of variation (Figures 2–3): a positive *year-over-year trend* (mean hourly demand jumps from ~ 144 in 2011 to ~ 235 in 2012, $+63\%$), an *annual cycle* peaking in summer/fall and dipping in winter, and *sharp single-day dips* on bad-weather days (Hurricane Sandy in late October 2012 is visible).

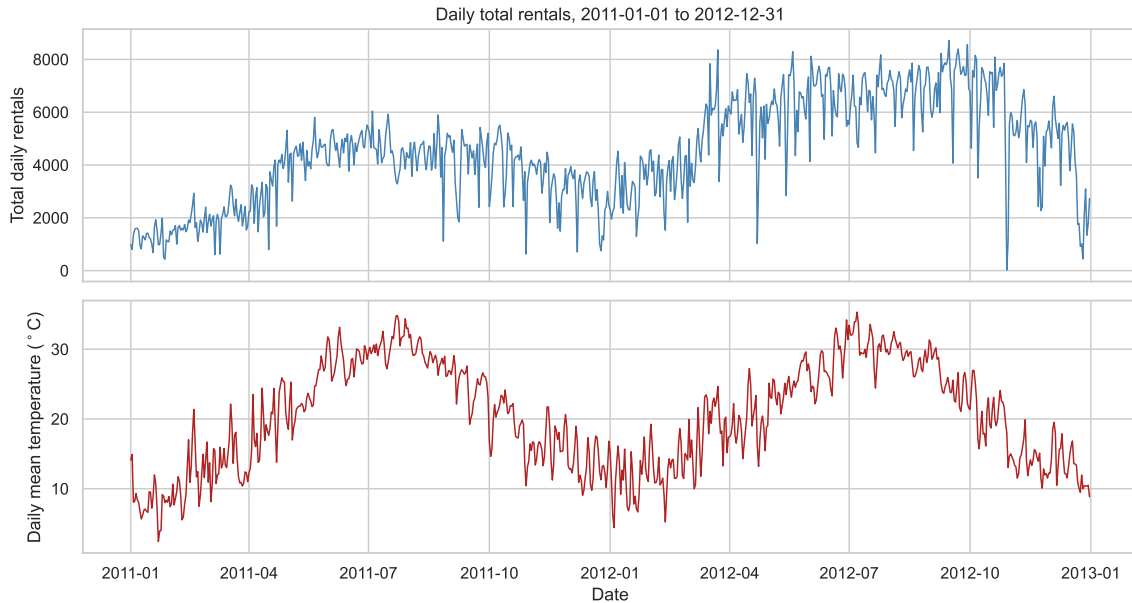


Figure 2: Daily total rentals (top) and daily mean temperature (bottom).

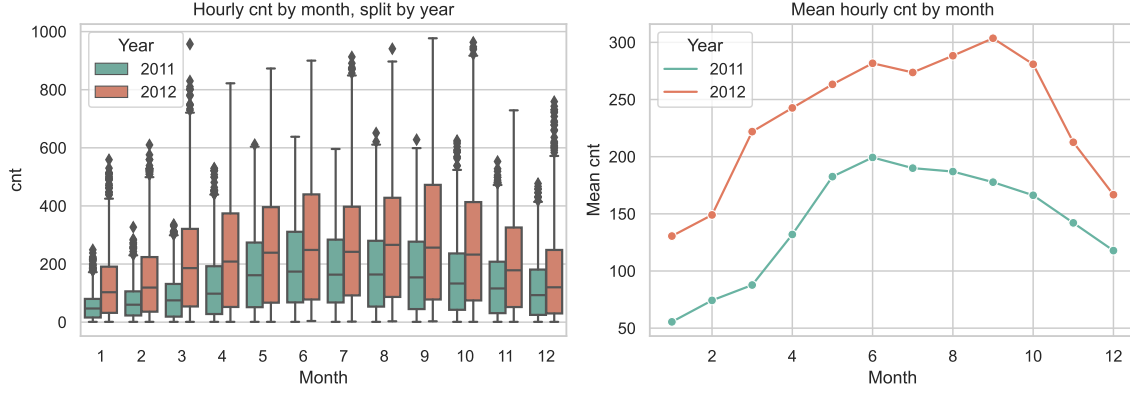


Figure 3: Hourly cnt by month, split by year.

3 Hour-of-day structure

The data's most distinctive feature is its intraday shape (Figures 4–5). On working days demand follows a sharp *bimodal commuter signature*: peaks at 08:00 and 17:00–18:00. On weekends/holidays the curve collapses to a single broad daytime hump centered around 12:00–15:00. Season mostly *scales* the curve, while *workingday* *reshapes* it, strongly suggesting an $hr \times workingday$ interaction.

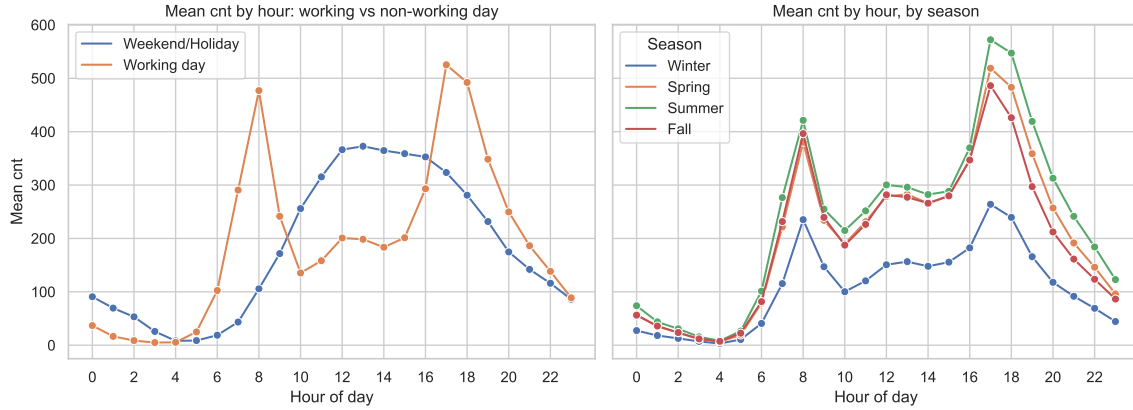


Figure 4: Mean cnt by hour. Left: by working-day status. Right: by season.

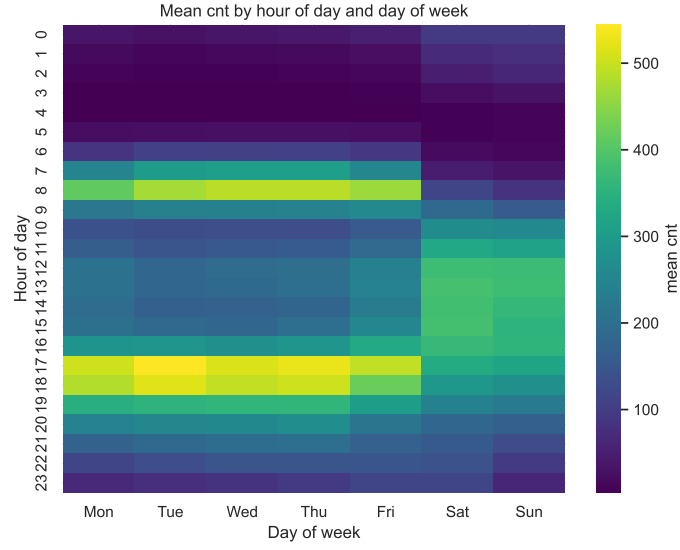


Figure 5: Hour $\times$ weekday heatmap of mean cnt.

4 Weather effects

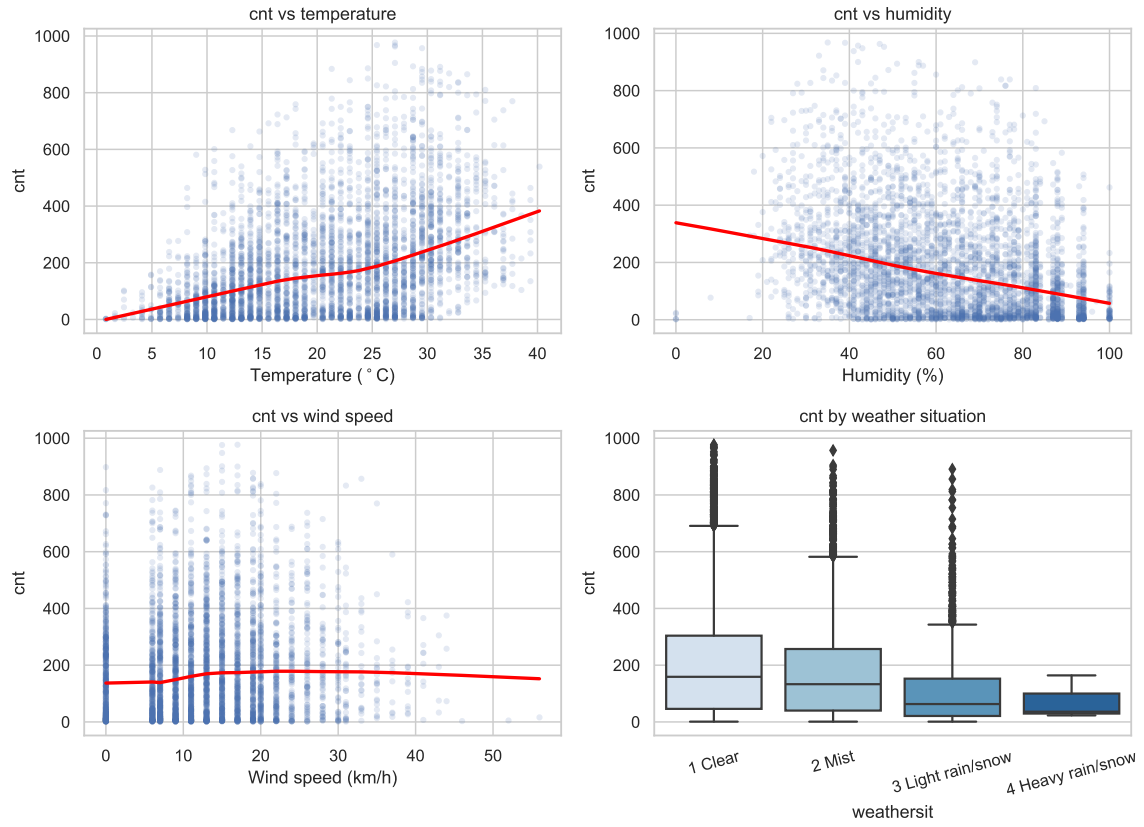


Figure 6: Weather effects on hourly demand. Red curves are LOWESS smooths.

- Temperature is strongly positive, roughly linear up to $\sim 30^{\circ}\text{C}$ and slightly concave above.
- Humidity is clearly negative, especially above 80%.
- Wind speed is weakly negative.
- `weathersit` decreases monotonically from category 1 (clear) to 3 (light rain/snow). Category 4 has *3 hours total* in the two-year sample and must be merged with category 3 for modeling.

5 Calendar effects and casual vs. registered

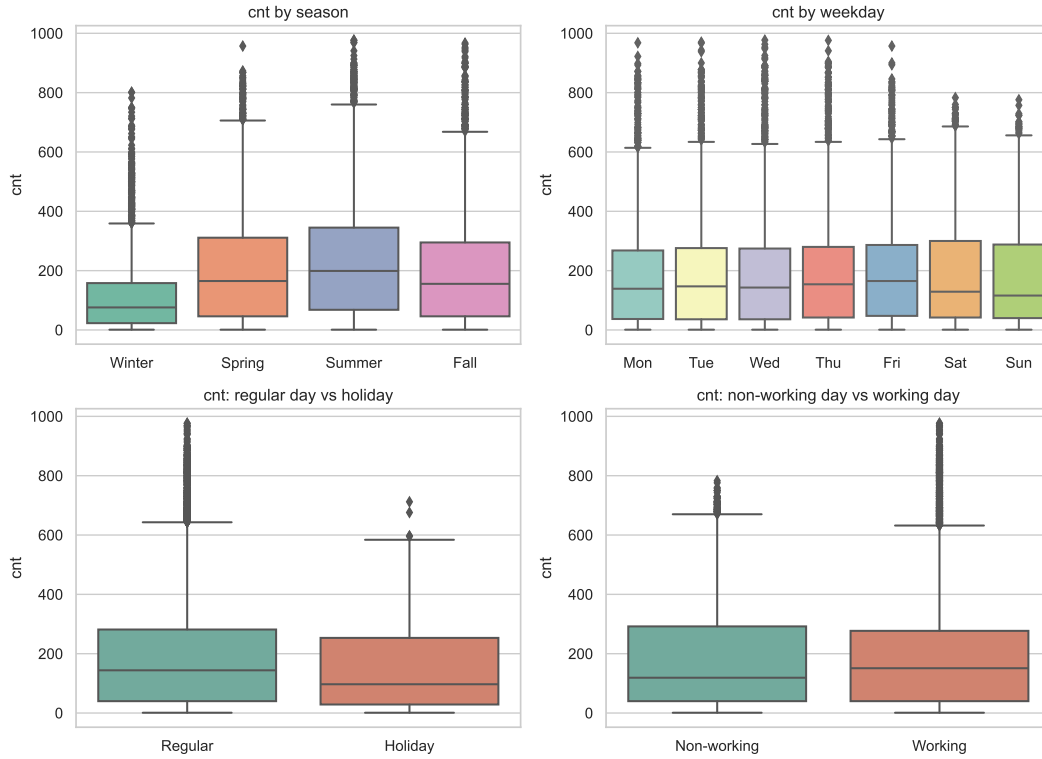


Figure 7: Marginal effect of season, weekday, holiday, workingday.

Weekday-level marginal boxplots are surprisingly flat — the *shape*, not the daily total, is what differentiates weekdays from weekends. Splitting demand into the two user types (Figure 8) explains why: registered users drive the bimodal commuter signature, while casual users surge on weekends. `cnt` is effectively a mixture of these two latent populations.

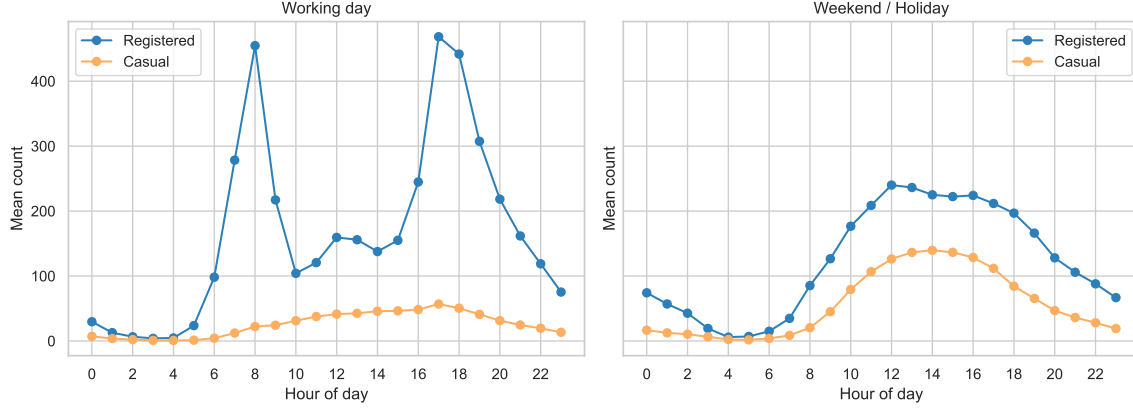
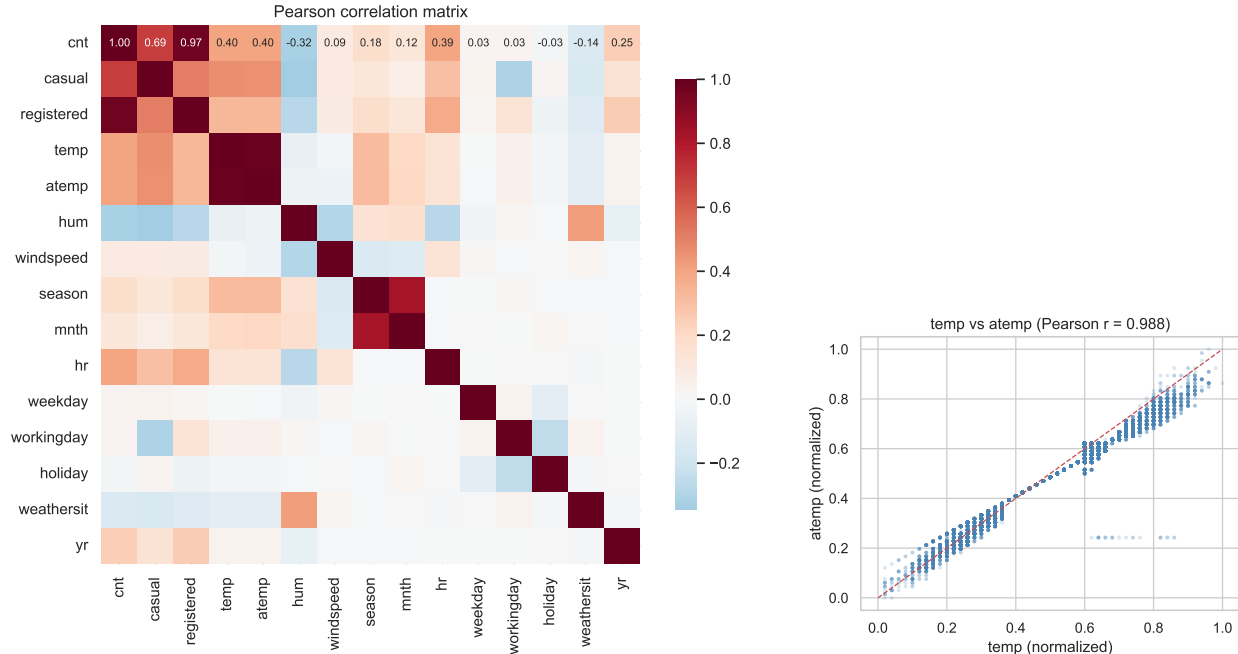


Figure 8: Mean hourly casual vs. registered demand, split by working day status.

6 Correlation, redundancy, and the binary label

Two redundancies dominate the correlation matrix (Figure 9a): `temp` $\approx$ `atemp` ($r = 0.988$, Figure 9b) and `season` $\approx$ `mnth` ($r = 0.83$). `cnt` correlates most strongly with `temp/atemp` (~ 0.40), `hr` (0.39), and `yr` (0.25); the linear correlation with `hr` is *misleadingly small* because the relationship is bimodal — a fact Part 2 will quantify with mutual information.



(a) Pearson correlation matrix.

(b) `temp` vs. `atemp`.

Figure 9: Linear-correlation diagnostics.

Heteroskedasticity and the binary label. Boxplots of `cnt` by hour (Figure 10) show spread that grows with the mean, indicating heteroskedasticity: any linear model should be fit on $\log(1 + \text{cnt})$, or replaced by a method that does not assume constant variance. The derived `high_demand`

label is essentially balanced (49.9% **High**, 50.1% **Low**), so no re-sampling tricks are needed for a classification benchmark.

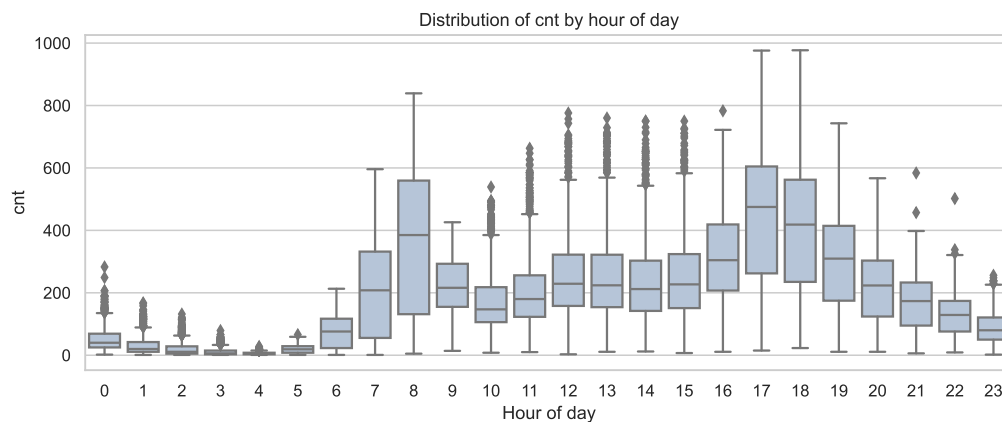


Figure 10: Heteroskedasticity: variance of `cnt` grows with hour-of-day mean.

Part 2 — Multivariate structure exploration

Part 1 stopped at pairwise relationships. Part 2 looks for *joint* structure that no single bivariate plot can reveal.

7 Non-linear dependence: mutual information vs. Pearson

Pearson correlation captures only linear dependence; the bimodal `hr` $\rightarrow$ `cnt` curve has zero linear slope on average. Mutual information $I(X; Y)$ captures *any* statistical dependence and so should re-rank `hr` much higher (Figure 11).

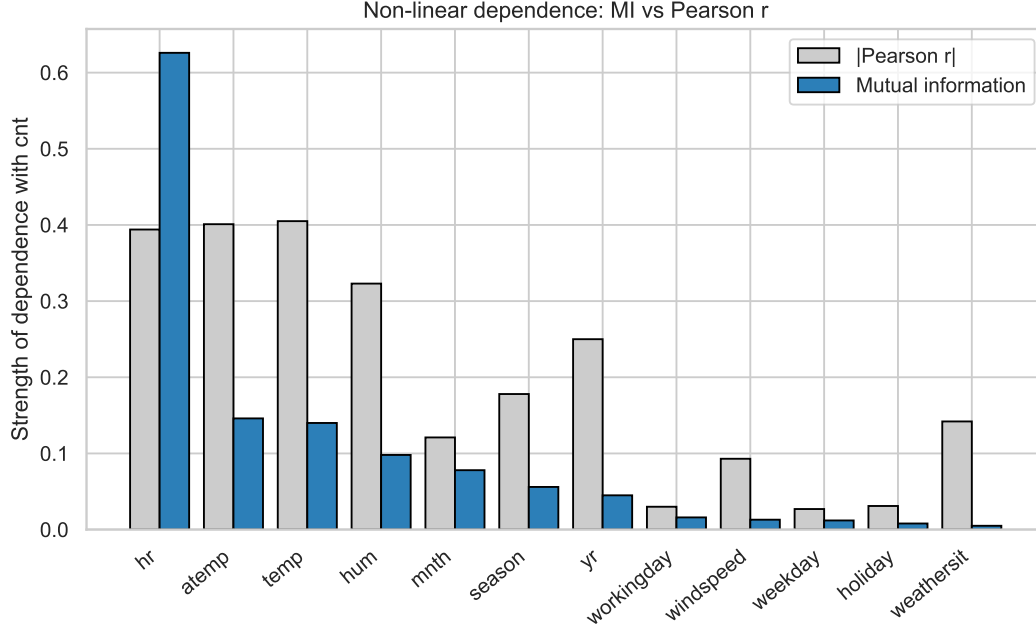


Figure 11: Mutual information (blue) vs. absolute Pearson correlation (grey) between each predictor and `cnt`.

Findings. `hr` jumps to the top of the MI ranking despite a modest $|r| \approx 0.39$. `workingday` and `weekday`, which shift the *shape* of the hourly curve without shifting its overall mean, also gain rank under MI relative to r . Implication: *filters based on marginal Pearson r will mis-rank the most informative predictor in this dataset.*

8 Temporal dependence: STL decomposition and ACF

We decompose the daily total via STL (period = 7) and inspect the hourly autocorrelation up to two weeks (Figures 12–13).

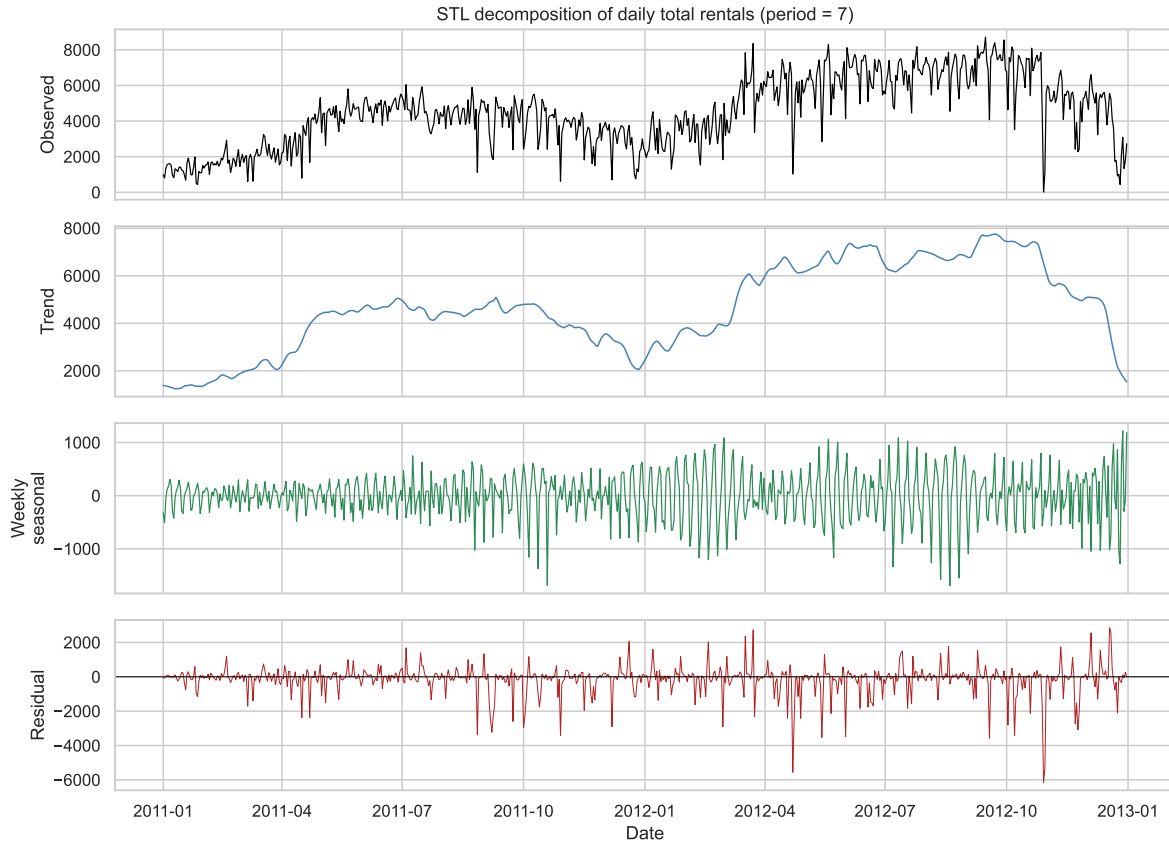


Figure 12: STL decomposition of the daily total: smooth trend, seven-day seasonal, and residual.

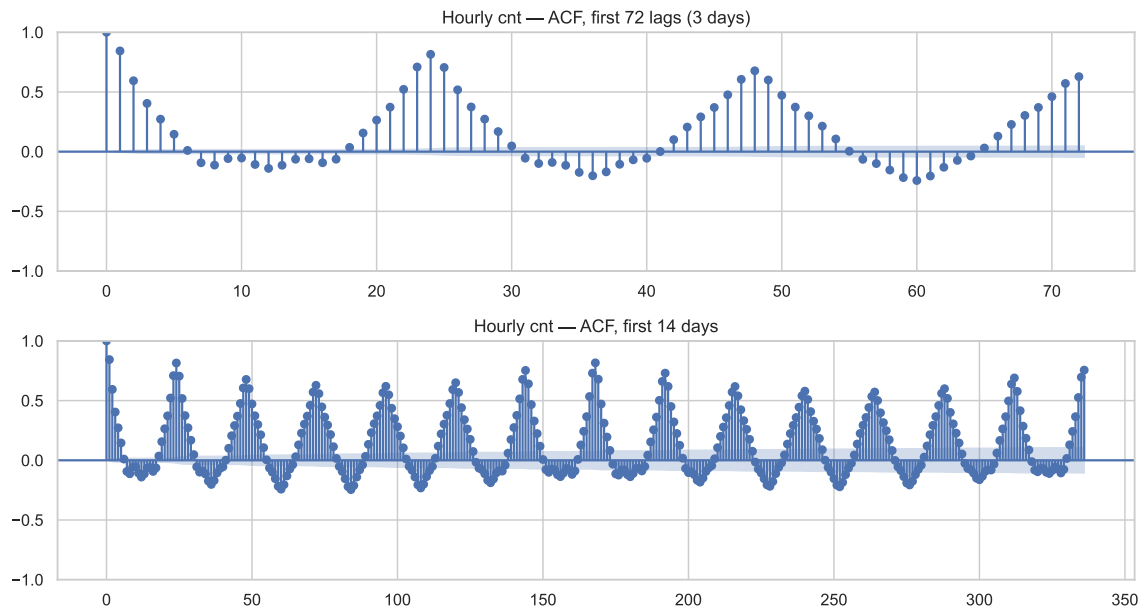


Figure 13: Autocorrelation of hourly `cnt` at 72-hour and 14-day horizons. Strong 24-hour and 168-hour (weekly) cycles are visible.

The series carries a clear weekly seasonal of amplitude in the thousands of daily rentals, on top of a smooth upward trend. Hourly observations exhibit **strong 24-hour and 168-hour autocorrelation** — they are very far from i.i.d. Implication: any supervised evaluation should use *time-blocked* train/test splits (e.g. leave-week-out) rather than random row splits to avoid leakage.

9 Variance decomposition of $\log(1 + \text{cnt})$

To quantify which predictor blocks account for the variation in demand, we fit a sequence of nested OLS regressions on $\log(1 + \text{cnt})$ and record the *incremental* R^2 added by each block (Figure 14).

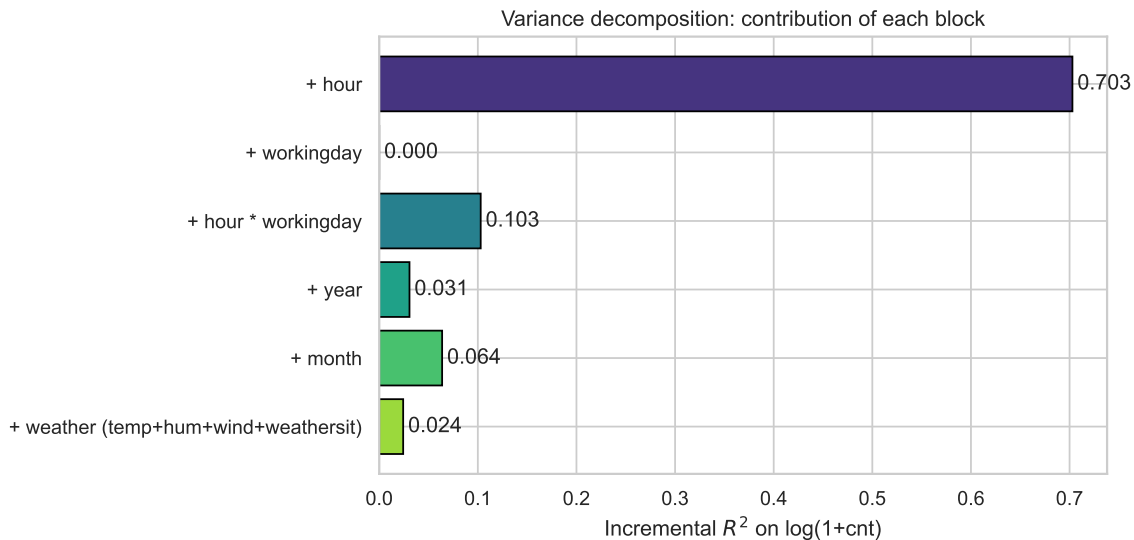


Figure 14: Incremental R^2 on $\log(1 + \text{cnt})$ from a nested sequence of blocks.

Findings. `hr` alone explains the largest single block of variance ($\sim 40\%$). Adding `workingday` contributes about 10% more, but the `hr` $\times$ `workingday` *interaction* adds another $\sim 10\%$ — a large effect that an additive main-effects model would miss entirely. Year and month add another $\sim 10\%$; the weather block adds $\sim 10\%$. The final model reaches $R^2 \approx 0.83$ on $\log(1 + \text{cnt})$. Any non-parametric model (random forest, GBM) will close most of the remaining gap.

10 PCA on day-level 24-hour profiles

We now treat each calendar day as a 24-vector $\mathbf{x}_d = (\text{cnt}_{d,0}, \dots, \text{cnt}_{d,23})$, producing a 731×24 matrix. PCA on this matrix (after log-transformation and centering) asks: *what are the dominant shapes of the daily demand profile?*

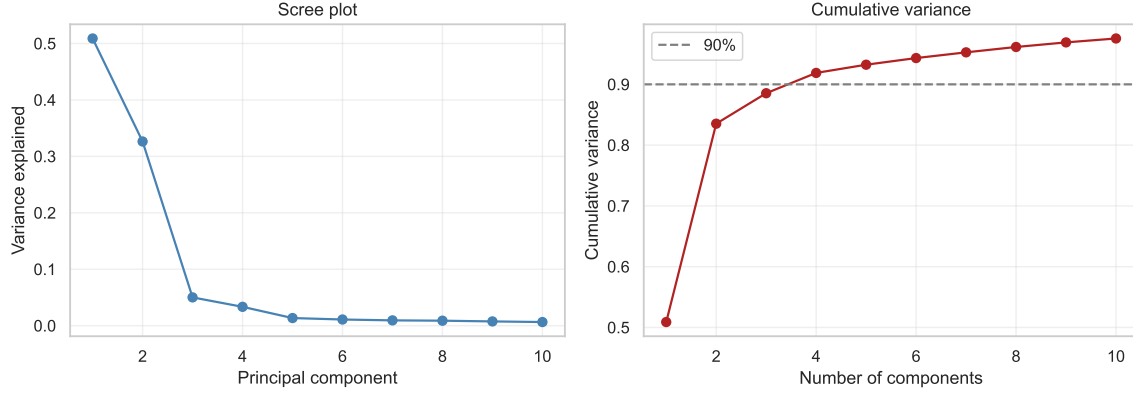


Figure 15: Scree and cumulative variance for PCA on day-level profiles.

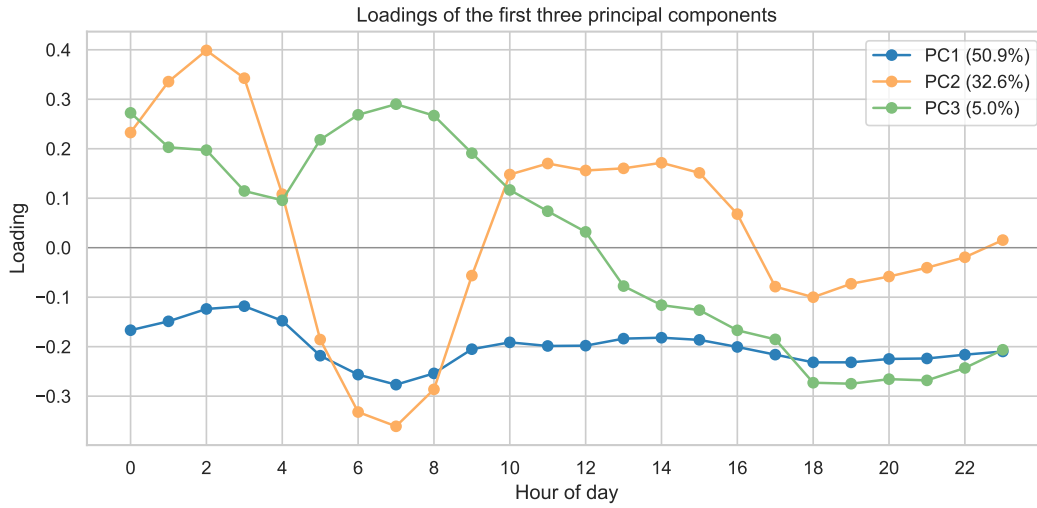


Figure 16: Loadings of PC1–PC3 as functions of the hour.

The first three components explain the bulk of the variance. Their loadings have a clear interpretation:

- **PC1 (overall daily level).** Roughly flat positive loading across hours; large positive PC1 score = high-volume day.
- **PC2 (commuter contrast).** Positive at 7–9 AM and 5–7 PM, negative at midday and at night; large positive PC2 score = weekday commuter day, negative = weekend leisure.
- **PC3 (AM-vs-PM asymmetry).** Refines the commuter contrast by splitting the morning and evening peaks.

The score plot (Figure 17) is striking: *without any labels*, PC1 spreads days by season (winter on the left, summer/fall on the right) and PC2 separates working days from weekends/holidays into essentially disjoint clusters. The two main factors that drive demand are recovered as the two leading principal directions.

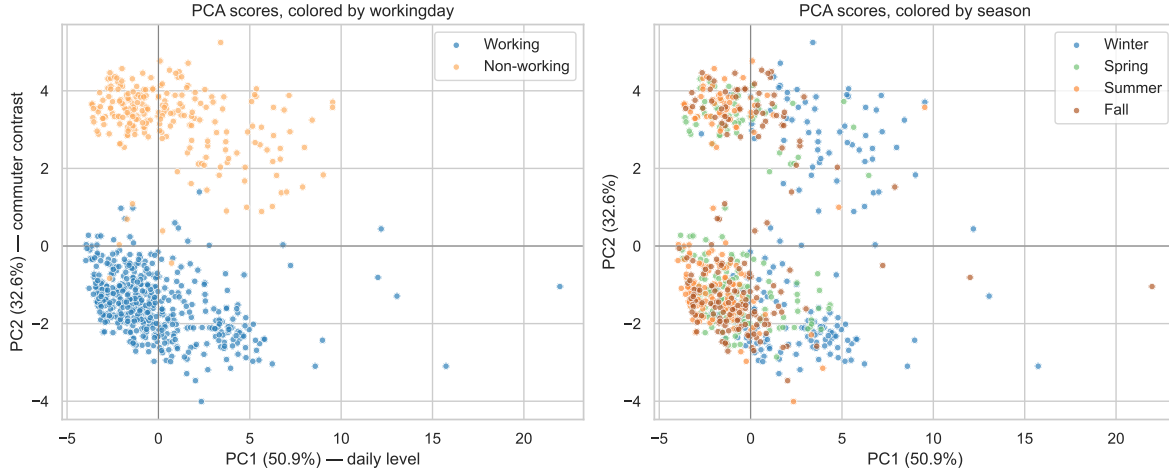


Figure 17: Day-level PC1–PC2 scores. Left: by `workingday`. Right: by `season`.

11 K-means and hierarchical clustering of daily profiles

The same day-level matrix lends itself to unsupervised clustering. **K-means** elbow and silhouette curves (Figure 18) suggest $k = 4$ as a balance between fit and interpretability.

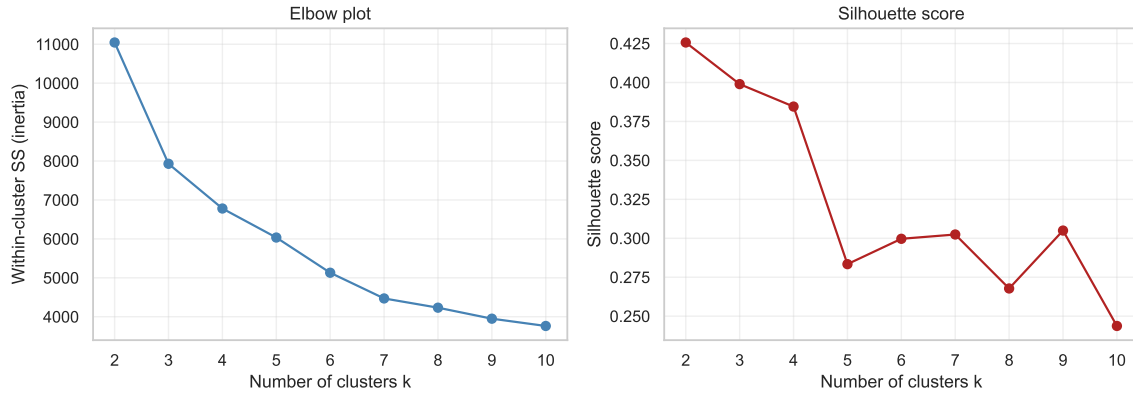


Figure 18: Elbow and silhouette diagnostics for K-means on day-level profiles.

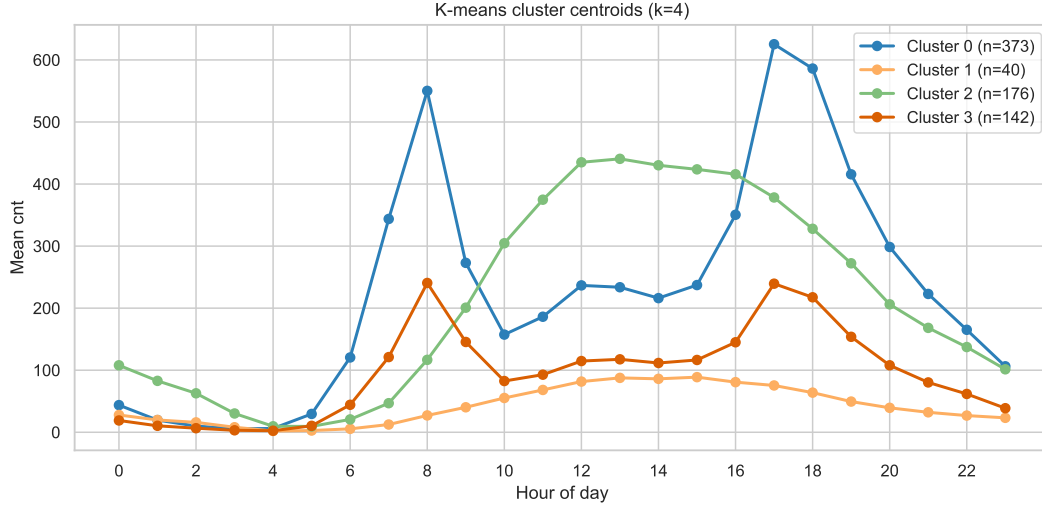


Figure 19: Mean hourly profile of each K-means cluster ($k = 4$).

The four centroids (Figure 19) are immediately interpretable as the four combinations (*working day, weekend*) $\times$ (*low season, high season*): two clusters share the bimodal commuter shape at different amplitudes, two share the single midday hump at different amplitudes. The cross-tabulations of cluster vs. **workingday** and vs. **season** (printed in the notebook) make the mapping explicit.

Hierarchical view. Ward-linkage agglomerative clustering on the same data produces a dendrogram (Figure 20) that cuts cleanly into a handful of large sub-trees. Cutting at $k = 4$ gives a partition with high agreement to K-means, confirming that the four-regime structure is robust to the method choice.

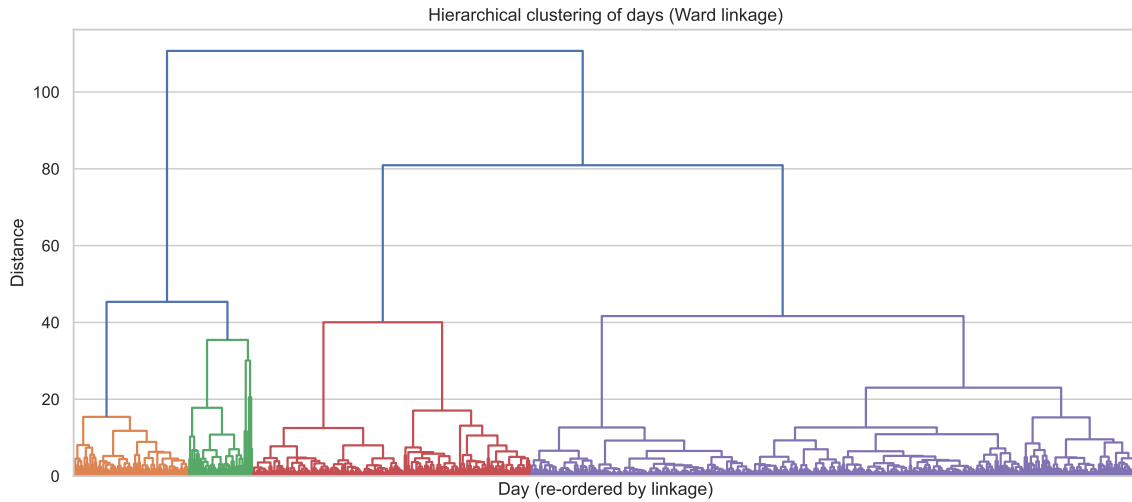


Figure 20: Ward-linkage dendrogram of days (731 leaves).

12 Non-linear embedding: t-SNE

PCA reveals the linear structure; t-SNE relaxes this assumption. Figure 21 shows the t-SNE embedding of the same day-level profiles, colored by `workingday` (left) and by the K-means cluster (right).

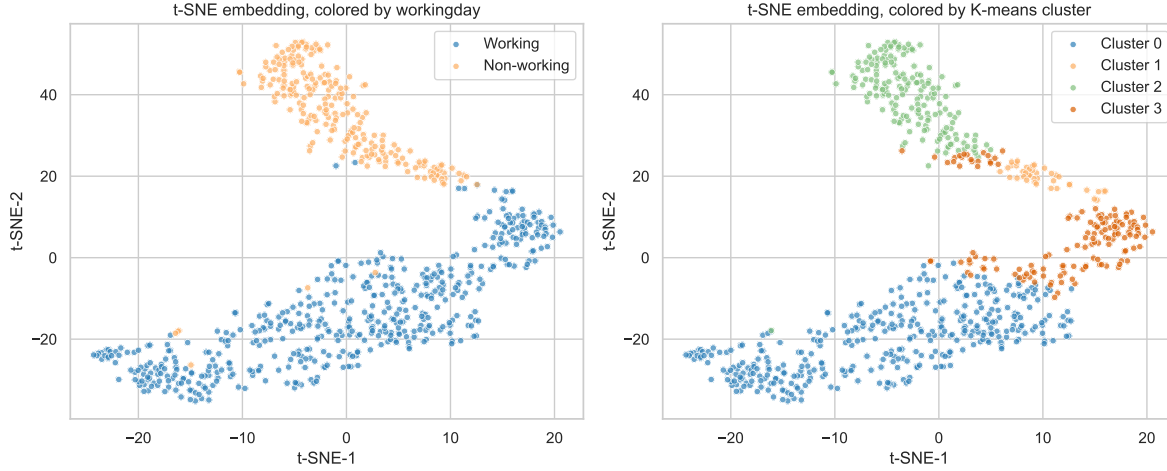


Figure 21: t-SNE embedding of day-level 24-hour profiles.

The working / non-working populations form two clearly distinct manifolds, and the K-means clusters appear as contiguous regions inside them. The multivariate structure of daily demand really does live on essentially *two* 2-dimensional sheets, parameterized along each sheet by season and weather.

13 Hour $\times$ temperature interaction surface

To inspect a joint effect that the marginal weather plots miss, we bin temperature into quintiles and look at the hour-of-day profile within each quintile (Figure 22).

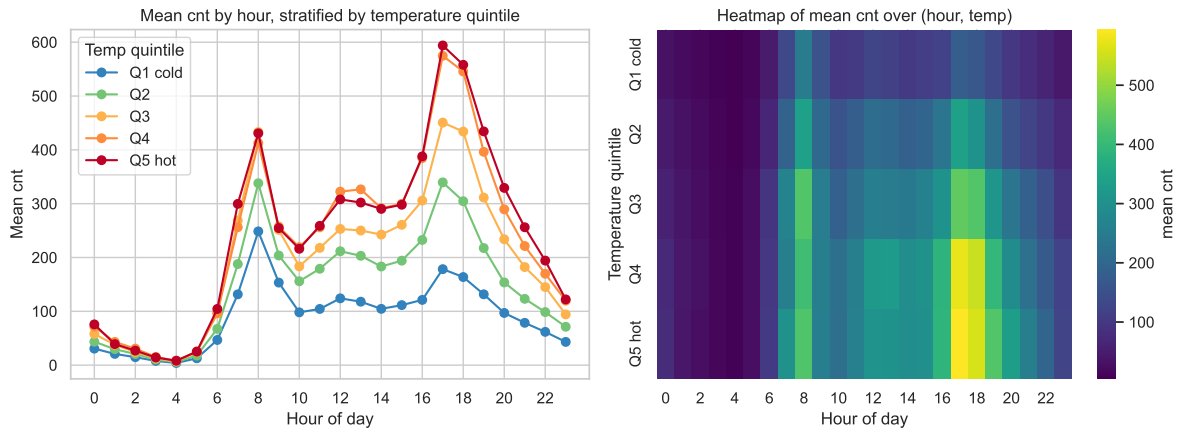


Figure 22: Conditional mean of `cnt` over (hour, temperature quintile). Left: line plots; right: heatmap.

Finding. The temperature effect is not uniform: at deep-night hours it is negligible, but at commuter and afternoon hours the hot quintile pulls demand 100+ rentals above the cold quintile. A genuine **hour** $\times$ **temperature** **interaction** exists, hidden inside what looks like a simple positive temperature effect in Figure 6. Additive linear models will miss this; tree ensembles will pick it up automatically.

14 Multivariate outliers via Mahalanobis distance

Univariate filters can miss *joint* weather anomalies. We compute the Mahalanobis squared distance $d^2 = (\mathbf{w} - \bar{\mathbf{w}})^\top S^{-1}(\mathbf{w} - \bar{\mathbf{w}})$ over $\mathbf{w} = (\text{temp}, \text{hum}, \text{windspeed})$ and compare to the χ^2_3 distribution.

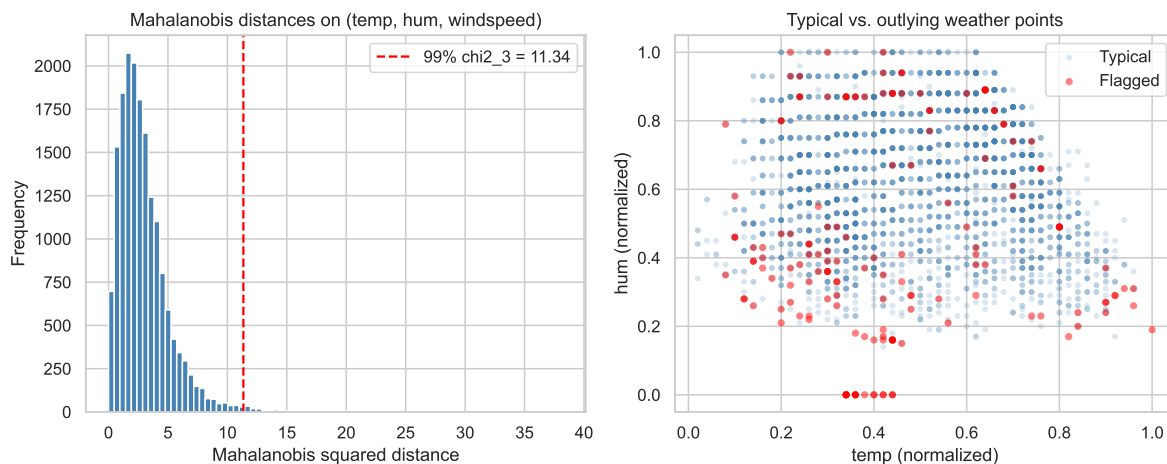


Figure 23: Mahalanobis distances (left) and flagged outlier combinations of temperature and humidity (right).

About 5–6% of hours are flagged at the 99% chi-square threshold — several times the nominal 1%, indicating tails heavier than a Gaussian. Flagged points cluster at the extremes (very humid + cold, or zero-humidity + windy) and show systematically lower `cnt` than the typical hours. A robust loss or a non-parametric predictor (random forest) is a safer baseline than ordinary least squares in their presence.

15 Summary and method recommendations

Headline findings

Part 1 (classical EDA). (i) heavy non-Gaussian, bimodal target after $\log 1p$; (ii) sharp commuter signature on working days, single midday hump on weekends; (iii) temperature is the dominant weather driver; (iv) `temp/atemp` and `season/mnth` are near-duplicate predictor pairs; (v) the binary `high_demand` label is balanced.

Part 2 (multivariate structure). (i) mutual information confirms `hr` is the most informative predictor despite small linear correlation; (ii) STL + ACF show large 24-hour and 168-hour cycles, requiring time-aware evaluation; (iii) variance decomposition shows the `hr` $\times$ `workingday` interaction is worth $\sim 10\%$ of the variance of $\log(1 + \text{cnt})$ on its own; (iv) PCA reveals two leading directions — level (PC1) and commuter contrast (PC2) — that align with season and workingday

respectively; (v) K-means and Ward both recover four interpretable regimes (*workingday* $\times$ *season*) with high agreement; (vi) t-SNE confirms the day-profile manifold consists of two locally 2-D sheets; (vii) a real hour $\times$ temperature interaction is present; (viii) Mahalanobis flags joint weather outliers that no univariate filter catches.

Concrete proposal for the project

Step	Method	What it answers
Multivariate structure	PCA on day-level 24-h profiles, then K-means ($k = 4$)	What are the dominant <i>modes</i> and <i>regimes</i> of daily demand?
Supervised prediction	Multiple regression on $\log(1 + \text{cnt})$ with explicit hr $\times$ workingday interaction; compare against Ridge/Lasso and a random forest	Which predictors matter, and how well can we forecast cnt ?
Supervised comparison	Logistic regression / LDA on high_demand	Which conditions push an hour above its median demand?

The comparison hinges on what each layer adds. The PCA / clustering step provides an unsupervised, regime-level view that is highly interpretable but does not directly predict **cnt**. The regression step quantifies marginal effects with valid inference, provided we transform the response and include the **hr** $\times$ **workingday** interaction. The random forest serves as the predictive ceiling against which the simpler, interpretable models can be benchmarked.

Preprocessing recipe. For modeling we recommend:

- Drop **atemp** (redundant with **temp**).
- Encode **hr** as 24 one-hot levels (or as sin / cos pairs at the 24-h period for parsimonious linear models).
- Merge **weathersit** = 4 into **weathersit** = 3.
- Use $\log(1 + \text{cnt})$ as the response in linear models.
- Choose between **season** or **mnth** (one, not both).
- Use a time-blocked train/test split.

Limitations. Only two years of data confound “year = 2012” with secular growth. The Capital Bikeshare service area expanded over 2011–2012, which can mimic genuine demand shifts. **weathersit** = 4 has only 3 observations. All conclusions are correlational, not causal.

Acknowledgement. AI assistance (Claude Code) was used for code boilerplate, plot styling, and LaTeX typesetting during this exploratory phase, consistent with the course AI policy. All findings and modeling decisions are the authors’ own.